

EXP 3

```
clc;
clear;
close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

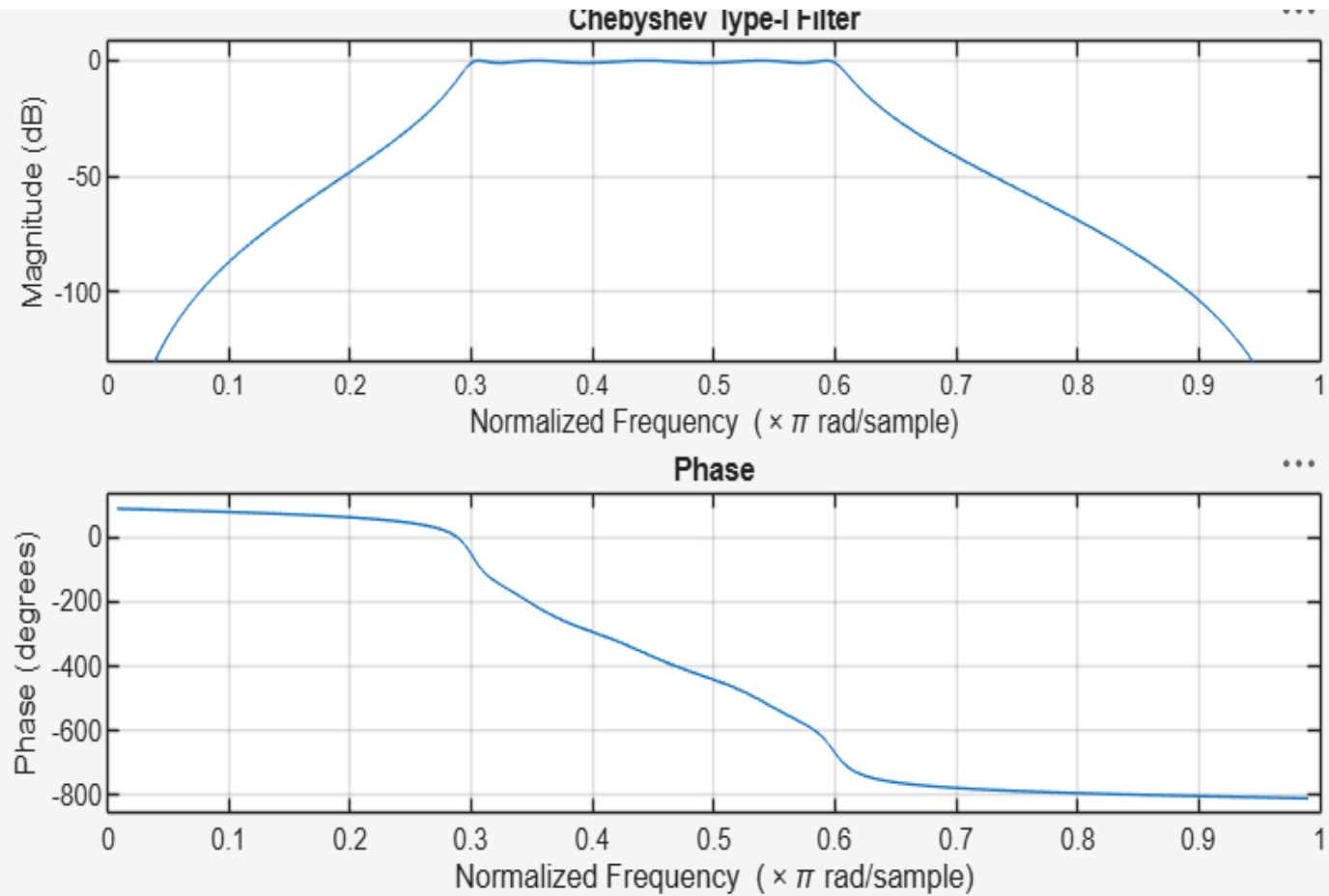
% Butterworth Filter
[n1,w1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,w1,'bandpass');

% Chebyshev Type-I Filter
[n2,w2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,w2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');
```



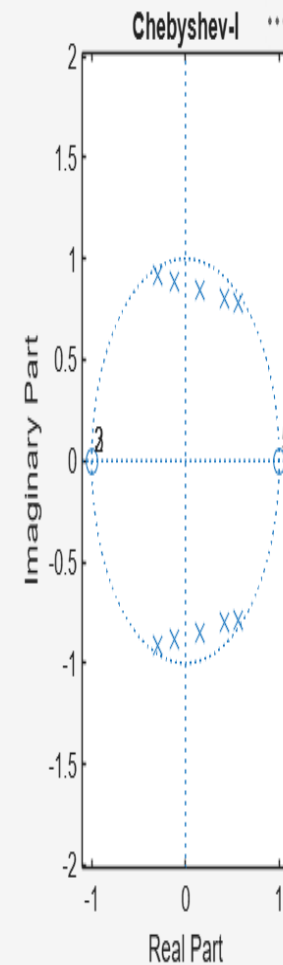
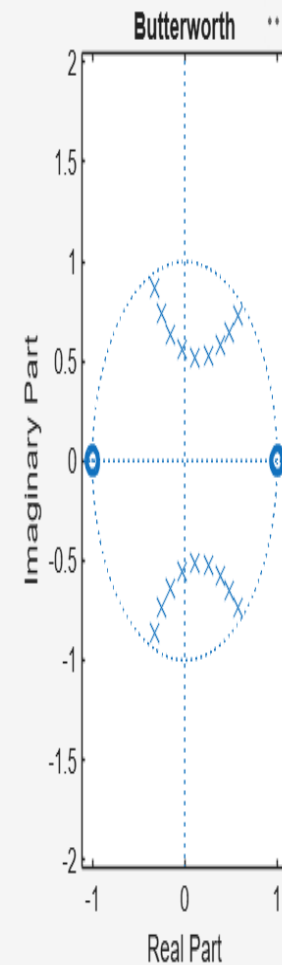
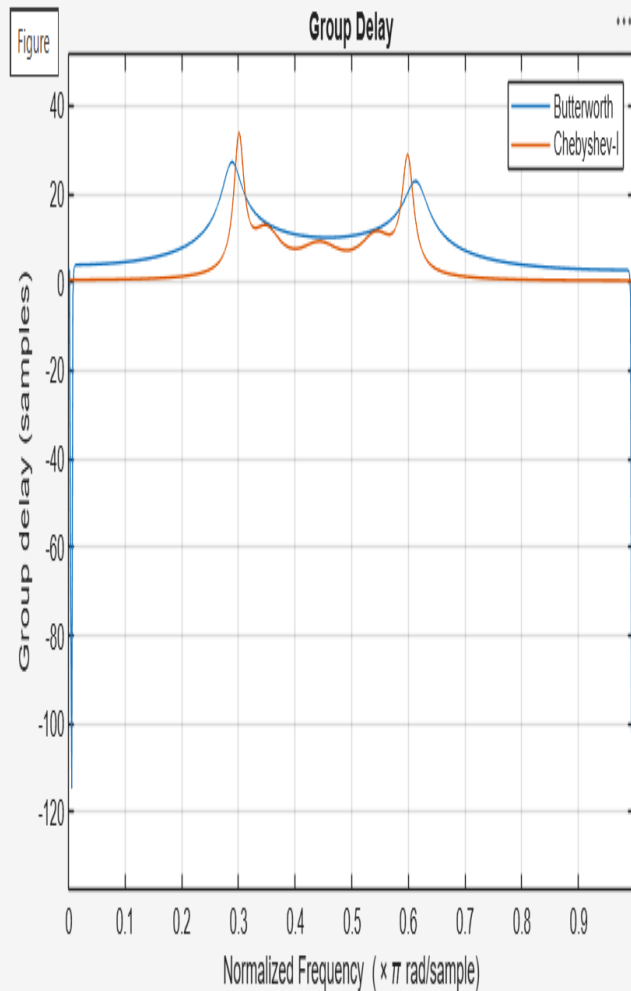
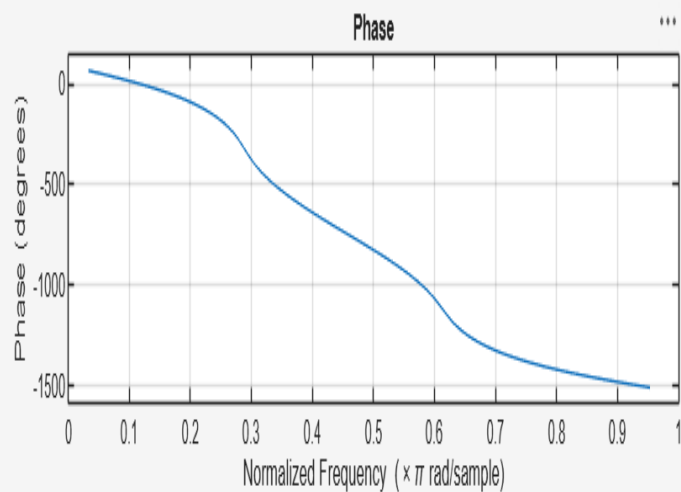
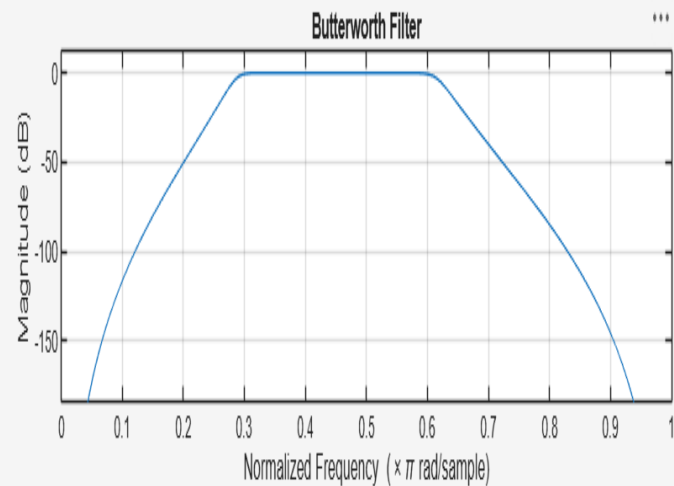


Figure 1 X Figure 2 X Figure 3 X Figure 4 X Figure 5 X Figure 6 X

